

(Morning Shift) ABES Engg. College, Ghaziabad
Solution of Pre-end Examination (2025-26)
 (25AS102) Fundamentals of Linear Algebra & Statistics

1. a)

Marks	No. of Students	Midpts (x_i)	$x_i - 25 = u$	$f \cdot u$
0-10	5	5	-2	-10
10-20	10	15	-1	-10
20-30	40	<u>25</u>	0	0
30-40	20	35	1	20
40-50	<u>25</u>	45	2	50
	<u>100</u>			<u>50</u>

$$\bar{x} = a + h \frac{\sum f \cdot u}{N} = 25 + 10 \left(\frac{50}{100} \right) = 30$$

$\frac{50}{100} \rightarrow 0.5$

(b)

Wages	No. of Workers	Cf
0-10	2	2
10-20	18	20
20-30	30	50
<u>30-40</u>	45	95
40-50	35	130
50-60	20	150
60-70	6	156
70-80	3	159

$$\frac{N}{2} = \frac{159}{2} = 79.5$$

Median class $\rightarrow$ 30-40

$$l = 30$$

$$h = 10$$

$$f = 45$$

$$C = 50$$

$$M_d = l + \frac{h}{f} \left(\frac{N}{2} - C \right)$$

$\frac{N}{2} - C \rightarrow 0.5$

$$= 30 + \frac{10}{45} (79.5 - 50) = 36.555$$

$\frac{10}{45} \rightarrow 0.5$

201 2 (a)

Marks	Students	Mid value	$f \times x^2$
1-3	15	2	60
3-5	18	4	288
5-7	27	6	972
7-9	10	8	640
9-11	6	10	600
	<u>76</u>		<u>2560</u>

$$\bar{x} = 5.315$$

$$\sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}} \quad \text{or} \quad \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2}$$

$$\sigma = \sqrt{\frac{2560}{76} - (5.31)^2} \approx 2.329$$

$(\text{Coeff of Variat}^n) = \frac{\sigma}{\bar{x}} \times 100 = 43.82\%$

No of patients	$x_i - \bar{x}$
153	1
147	-5
151	-1
156	4
153	1

$$\bar{x} = \frac{153+147+151+156+153}{5} = 152$$

$$\text{Mean deviation} = \frac{\sum |x_i - \bar{x}|}{N} = \frac{12}{5} = 2.4$$

$\approx 3 \text{ patients (approx)}$

$$\text{Coeffi of M.D} = \frac{\text{M.D}}{\bar{x}} = \frac{2.4}{152} = 0.0157 \text{ Any.}$$

20/3 (a)

x	y	x^2	x^3	x^4	xy	x^2y
1	6	1	1	1	6	6
2	11	4	8	16	22	44
3	18	9	27	81	54	162
4	27	16	64	256	108	432
<u>10</u>	<u>62</u>	<u>30</u>	<u>100</u>	<u>354</u>	<u>190</u>	<u>644</u>

The general eqnⁿ of second degree is

$$y = a + bx + cx^2 \quad \text{--- (0.5)}$$

The corresponding normal equations are

$$\sum y = na + b\sum x + c\sum x^2 \Rightarrow 62 = 4a + 10b + 30c$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3 \Rightarrow 190 = 10a + 30b + 100c$$

$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4 \Rightarrow 644 = 30a + 100b + 354c \quad \text{--- (1)}$$

$$a = 3, b = 2, c = 1$$

$$y = 3 + 2x + x^2 \quad \text{--- (0.5)}$$

(b) (i) $y = ax + \frac{b}{\sqrt{x}}$

$$a\sum x^2 + b\sum \sqrt{x} = \sum xy$$

$$a\sum \sqrt{x} + b\sum \left(\frac{1}{\sqrt{x}}\right) = \sum \left(\frac{y}{\sqrt{x}}\right) \quad \text{--- (1)}$$

(ii) $y = \frac{c_0}{x} + c_1 x^2$

$$\sum x^2y = c_1\sum x^4 + c_0\sum x$$

$$\sum \frac{y}{x} = c_1\sum x + c_0\sum \frac{1}{x^2} \quad \text{--- (1)}$$

(iii) $y = a_1 e^{-2t} + a_2 e^{-5t}$

$$\sum y e^{-2t} = a_1 \sum e^{-4t} + a_2 \sum e^{-7t}$$

$$\sum y e^{-5t} = a_1 \sum e^{-7t} + a_2 \sum e^{-10t} \quad \text{--- (1)}$$

Q.N. - 4(a)

(4)

X	68	64	75	50	64	80	75	40	55	64	Total
Y	62	58	68	45	81	60	68	48	50	70	
R _x	4	6	2.5	9	6	1	2.5	10	8	6	
R _y	5	7	3.5	10	1	6	3.5	9	8	2	
D = R _x - R _y	-1	-1	-1	-1	5	-5	-1	1	0	4	
D ²	1	1	1	1	25	25	1	1	0	16	$\Sigma D^2 = 72$

Here $n = 10$ In X series : (i) 75 occurs twice $\Rightarrow m_1 = 2$

$$\text{tie Rank} = \frac{2+3}{2} = 2.5$$

(ii) 64 occurs thrice $\Rightarrow m_2 = 3$

$$\text{tie Rank} = \frac{5+6+7}{3} = 6$$

In Y series : (i) 68 occurs twice $\Rightarrow m_3 = 2$

$$\text{tie Rank} = \frac{3+4}{2} = 3.5$$

 $\therefore$ Rank Correlation

$$r = 1 - \frac{6 \left\{ \Sigma D^2 + \frac{1}{12} m_1 (m_1^2 - 1) + \frac{1}{12} m_2 (m_2^2 - 1) + \frac{1}{12} m_3 (m_3^2 - 1) \right\}}{n (n^2 - 1)}$$

$$= 1 - \frac{6 \left\{ 72 + \frac{1}{12} \times 2 (2^2 - 1) + \frac{1}{12} \times 3 (3^2 - 1) + \frac{1}{12} \times 2 (2^2 - 1) \right\}}{10 (10^2 - 1)}$$

$$= 1 - \frac{6 \times 75}{990} = \frac{6}{11}$$

$$= 0.545$$

(1)

Q.N. - 4(b)

(a) $\therefore$ Both reg. lines pass through $(\bar{x}, \bar{y})$, therefore

$$8\bar{x} - 10\bar{y} + 66 = 0 \quad \text{and} \quad 40\bar{x} - 18\bar{y} - 214 = 0$$

Solving these two equations, we have

$$\boxed{\bar{x} = 13, \bar{y} = 17}$$

— (1)

(b) Variance of $x = \sigma_x^2 = 9 \Rightarrow \sigma_x = 3$

Rewriting the eqn of reg. lines as

$$y = 0.8x + 6.6 \quad \text{and} \quad x = 0.45y + 5.35$$

we get, $b_{yx} = 0.8$ and $b_{xy} = 0.45$

$$\therefore r^2 = b_{xy} \cdot b_{yx} = 0.45 \times 0.8 = 0.36$$

(c) $\Rightarrow \boxed{r = 0.6}$

— (1)

$$\therefore b_{yx} = r \frac{\sigma_y}{\sigma_x} \Rightarrow 0.8 = 0.6 \times \frac{\sigma_y}{3}$$

(b) $\Rightarrow \sigma_y = \frac{0.8 \times 3}{0.6} = 4 \Rightarrow \boxed{\sigma_y = 4}$ — (1)

Q.N. 5(a)

Class	Freq	x	fx	$f(x-\bar{x})^2$	$f(x-\bar{x})^4$
0-10	8	5	40	6361.92	5059253.261
10-20	12	15	180	3974.88	1316639.251
20-30	20	25	500	1344.8	90424.352
30-40	30	35	1050	97.2	314.928
40-50	15	45	675	2088.6	290816.664
50-60	10	55	550	4752.4	2258530.576
60-70	5	65	325	5056.2	5113031.688
	$\Sigma f = 100$		$\Sigma fx = 3320$	$\Sigma f(x-\bar{x})^2 = 23676$	$\Sigma f(x-\bar{x})^4 = 14129010.72$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{3320}{100} = 33.2 \quad \text{— (2)}$$

$$\text{Measure of kurtosis} = \frac{\mu_4}{\mu_2^2} \quad \text{--- } (0.5)$$

$$\mu_2 = \frac{\sum f(x - \bar{x})^2}{\sum f} = \frac{23676}{100} = 236.76 \quad \text{--- } (0.5)$$

$$\mu_4 = \frac{\sum f(x - \bar{x})^4}{\sum f} = \frac{14129010.72}{100} = 141290.1072 \quad \text{--- } (0.5)$$

$$\text{Measure of kurtosis} = 2.52 < 3 \text{ (Platykurtic)} \quad \text{--- } (0.5)$$

Q.N. 5(b)

We have $A = 2$, $\mu'_1 = 1$, $\mu'_2 = 16$, $\mu'_3 = -40$, $\mu'_4 = 100$

$$\mu_4 = 0$$

$$\mu_2 = \mu'_2 - \mu_1'^2 = 16 - 1^2 = 15$$

$$\begin{aligned} \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 \\ &= -40 - 3 \times 16 \times 1 + 2 \times 1^3 \\ &= -40 - 48 + 2 \\ &= -86 \end{aligned}$$

$$\begin{aligned} \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4 \\ &= 100 - 4(-40)(1) + 6(16)(1)^2 - 3(1)^4 \\ &= 100 + 160 + 96 - 3 \\ &= 353 \end{aligned} \quad \text{--- } (2)$$

$$\mu_1 = \bar{x} = A + \mu'_1 = 2 + 1 = 3$$

$$\mu_2 = \mu_2 + \bar{x}^2 = 15 + 3^2 = 24$$

$$\mu_3 = \mu_3 + 3\mu_2\bar{x} + \bar{x}^3 = -86 + 3(15)(3) + (3)^3 = 16$$

$$\begin{aligned} \mu_4 &= \mu_4 + 4\mu_3\bar{x} + 6\mu_2\bar{x}^2 + \bar{x}^4 \\ &= 353 + 4(-86)(3) + 6(15)(3)^2 + (3)^4 \\ &= 212 \end{aligned} \quad \text{--- } (2)$$

Q.N. - 6(a)

(7)

$$y = a e^{bx} \Rightarrow \log_{10} y = \log_{10} a + x \cdot b \log_{10} e \Rightarrow y = A + x \cdot B$$

Reqd. Normal eqns.

$$\text{where } Y = \log_{10} y$$

$$A = \log_{10} a$$

$$B = b \log_{10} e$$

$$\Sigma Y = nA + B \Sigma x$$

$$\Sigma xY = A \Sigma x + B \Sigma x^2 \quad \text{--- (1)}$$

x	y	Y = log ₁₀ y	x ²	xY
1	10	1.0000	1	1
5	15	1.1761	25	5.8805
7	12	1.0792	49	7.5544
9	15	1.1761	81	10.5849
12	21	1.3222	144	15.8664
<u>Σx = 34</u>		<u>ΣY = 5.7536</u>	<u>Σx² = 300</u>	<u>ΣxY = 40.8862</u>

Substituting these values in N.E. --- (2)

$$5.7536 = 5A + 34B$$

$$40.8862 = 34A + 300B$$

On solving, we get

$$A = 0.9766 \Rightarrow a = \text{Antilog}_{10} A = 9.4754$$

$$B = 0.02561 \Rightarrow b = \frac{B}{\log_{10} e} = 0.059$$

Hence reqd. curve is $y = 9.4754 e^{0.059x}$ --- (1)

Q.N. - 6(b)

x	y	xy	x ²
1	15	15	1
3	18	54	9
5	21	105	25
7	23	161	49
19	22	418	361
<u>35 (Σx)</u>	<u>99 (Σy)</u>	<u>753 (Σxy)</u>	<u>445 (Σx²)</u>

--- (1)

Here $n = 5$

$$\bar{x} = \frac{\sum x}{n} = \frac{35}{5} = 7, \quad \bar{y} = \frac{\sum y}{n} = \frac{99}{5} = 19.8$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = 0.3 \quad \text{--- (1.5)}$$

Reg. line y on x is

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$(y - 19.8) = 0.3(x - 7)$$

$$y = 0.3x + 17.7 \quad \text{--- (1)}$$

When $x = 4$ years

$$y = 0.3 \times 4 + 17.7$$

$$= 18.90 \text{ hundred rupees}$$

$$= ₹ 1890 \quad \text{--- (0.5)}$$